

EXECUTIVE SUMMARY

Short-Term Ferry Ticket Demand Forecasting & Predictive Decision Support System

Prepared for Toronto Government Parks, Forestry & Recreation — Unified Mentor Internship Program

Toronto Island Park's ferry operations have historically relied on after-the-fact reporting: staff learn about a surge in demand only once it is already happening. This project turned ten years of ticket sales and redemption records into a working forecasting system that predicts demand 15 minutes to 2 hours in advance — giving operations staff real advance notice instead of a rear-view mirror.

10.6

Years of ticket data analyzed

15m-2hr

Forecast horizons covered

45%

More accurate than naive guessing at 2 hrs out

~87%

Of actual demand values fell inside the forecast range

What Was Built

A machine learning model was trained on 373,189 fifteen-minute intervals of cleaned historical data and evaluated against simpler forecasting methods. The best-performing approach, Gradient Boosting, was paired with an uncertainty range (not just a single guess, but a realistic “likely between X and Y” band) and delivered through an interactive dashboard where staff can select any date, time, and forecast horizon, compare models side by side, and see predicted vs. actual demand.

Key Findings

- Demand is highly predictable from recent patterns — primarily how busy the last hour has been — which the model uses to anticipate what comes next.
- The forecasting model's advantage over simple guessing grows the further ahead it looks: about 20% more accurate at 15 minutes out, but 45% more accurate at 2 hours out — exactly the lead time most useful for staffing decisions.
- Demand is strongly seasonal (August sees over 26 times the ticket volume of January) and strongly time-of-day driven (activity peaks near midday and is negligible overnight), both of which the model accounts for automatically.

- The system's uncertainty estimates are realistic and self-adjusting: forecast ranges are narrow (a few tickets) on predictable quiet days and automatically widen (100+ tickets) on high-demand days such as holiday weekends.

Recommendations

- Adopt the forecasting dashboard as a planning aid for staffing and crowd-management decisions, particularly using the 1-2 hour forecast window.
- Treat a widening forecast range as an early warning sign on its own — it flags rising uncertainty even before the demand number itself looks unusual.
- Refresh the model periodically as new data accumulates, and revisit it after any major change to schedules, pricing, or park programming.
- Consider flagging known high-demand dates (e.g. Canada Day, major festivals) so the system can account for them explicitly rather than treating them as ordinary days.

Next Steps

- Connect the dashboard to a live ticketing feed so forecasts reflect real-time conditions rather than historical backtests.
- Evaluate two additional forecasting techniques named in the original project scope (ARIMA/SARIMA and Prophet), which could not be tested in the current development environment.
- Pilot the dashboard with operations staff for one season to gather feedback before a fuller rollout.

Full technical methodology, data tables, and figures are documented in the accompanying research paper.